

Machine Learning

ePDG@CMInDS

Instructors: Sharayu Moharir & Jayakrishnan Nair

sharayum@ee.iitb.ac.in, jayakrishnan.nair@ee.iitb.ac.in

Lectures:

Monday (7.30 – 9.00 pm)

Saturday (9.30 – 11.00 am)

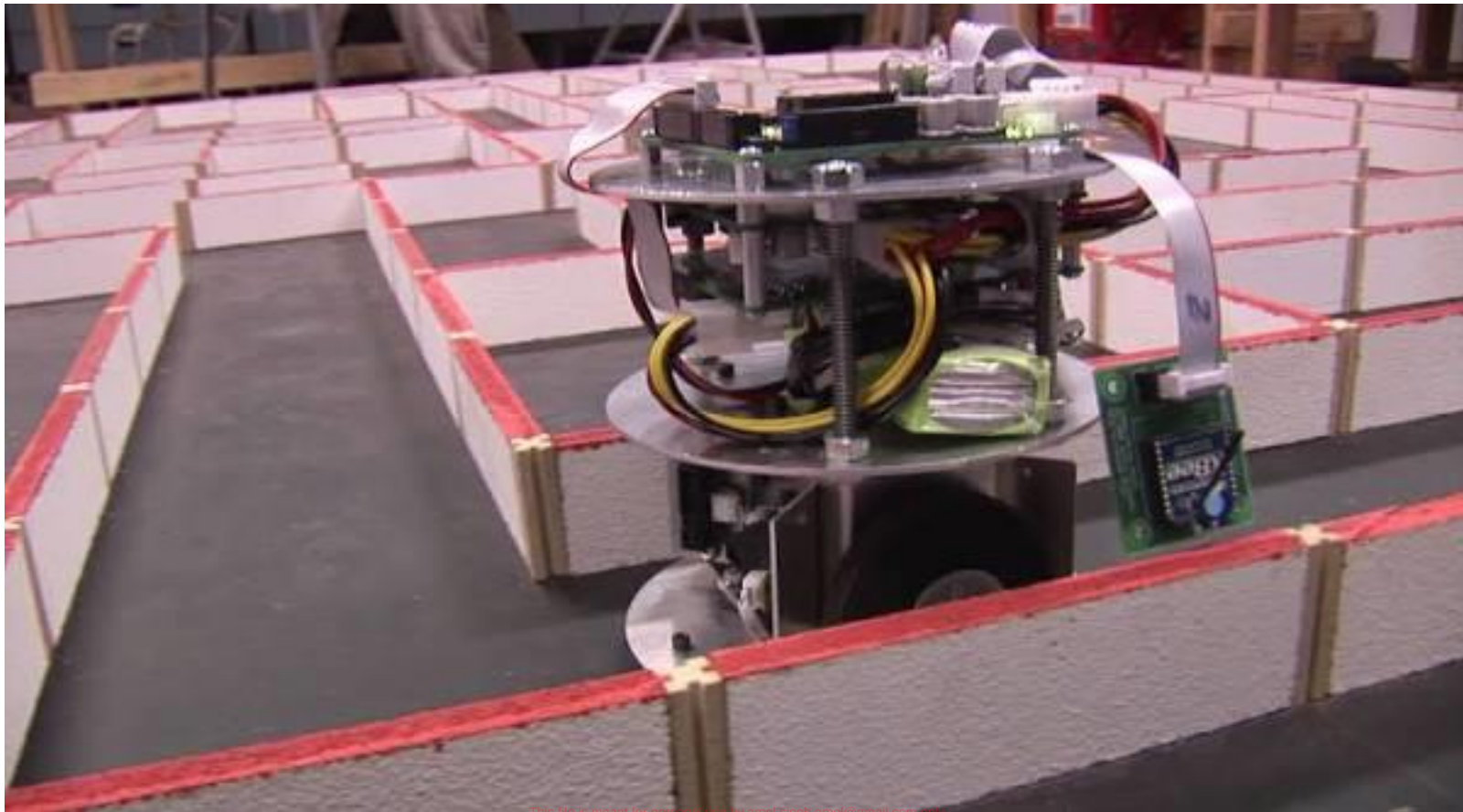
TA session: Saturdays (6.30 – 7.30 pm)

What is machine learning?

Can you write a computer program to ...
play tic-tac-toe?



Can you write a computer program to ...
get a robot to navigate a maze?



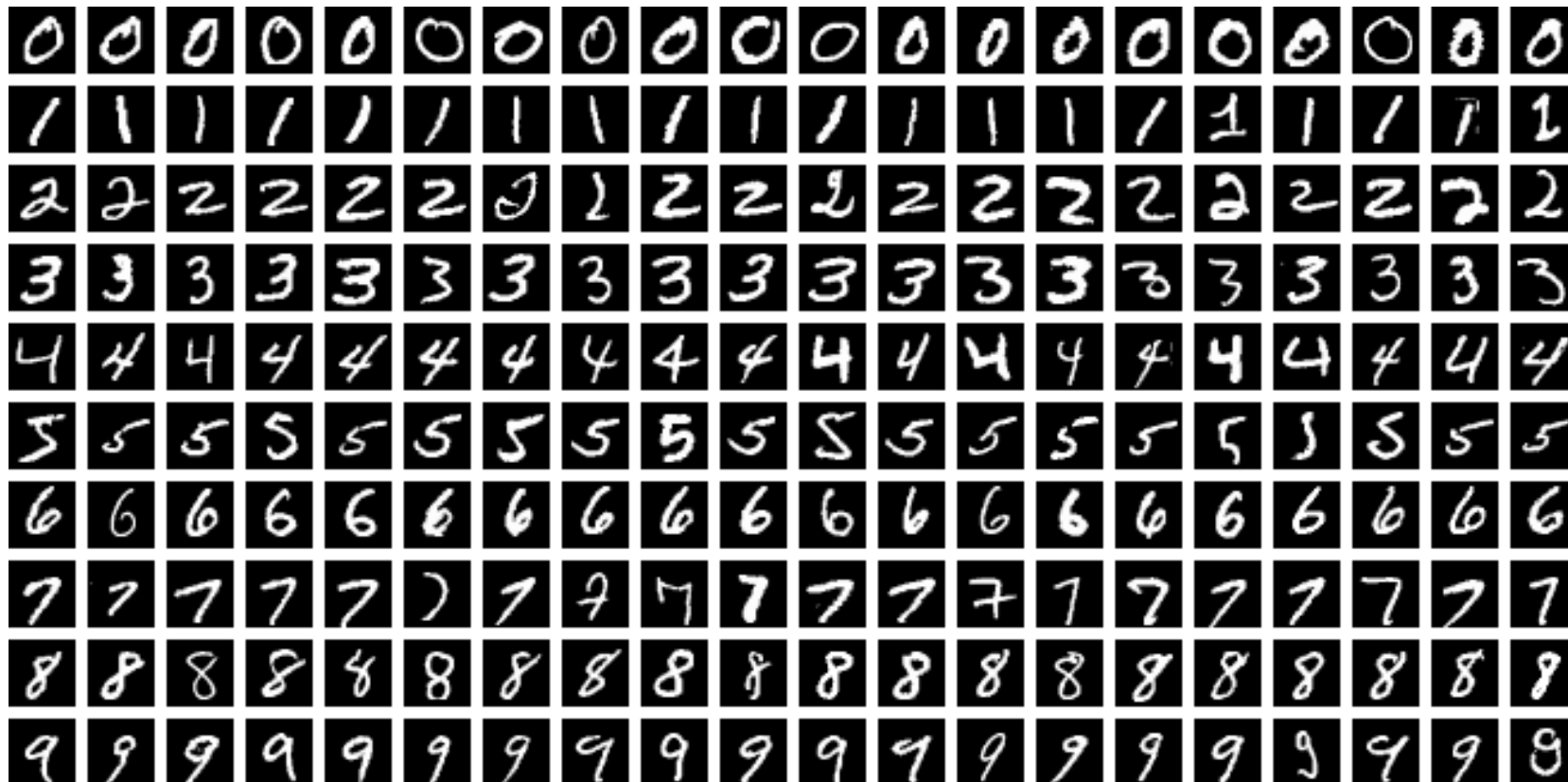
Can you write a computer program to ...

beat



at chess?

Can you write a computer program to ...
identify these handwritten digits?



Some basic definitions

Artificial intelligence:

Ability of computer systems and machines to perform tasks that conventionally require human intelligence.

Machine Learning:

Subset of AI that seeks to design systems that can learn from data to perform tasks without explicit programming.

Some basic definitions

Artificial intelligence:

Ability of computer systems and machines to perform tasks that conventionally require human intelligence.

Machine Learning:

Subset of AI that seeks to design systems that can learn from data to perform tasks without explicit programming.

millions of chess game records
labeled handwritten image data

Basic requirement for ML: *pattern* that can be learned from *data*

"In God we trust. All others must bring data."
W. Edward Demmings

Q: When does ML make sense?

There is a pattern to be learnt from data

The pattern is hard to define (and program explicitly)

E.g.: clean governing equations/laws unavailable

There is enough data about the pattern

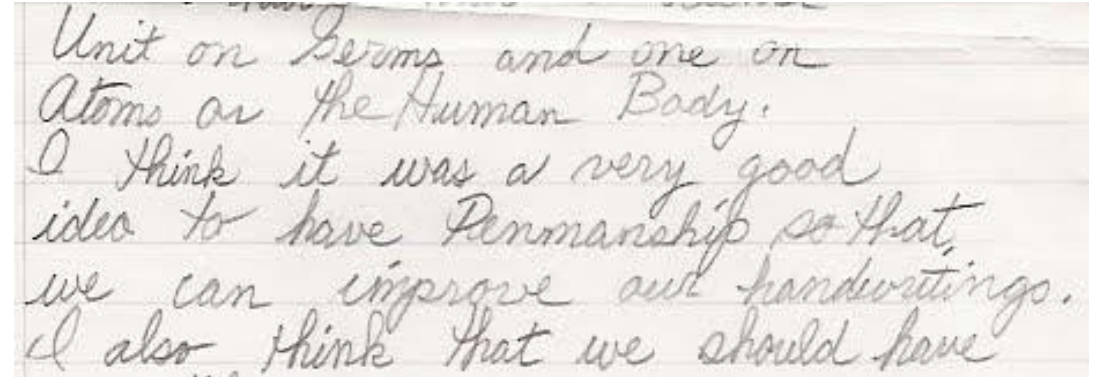
Which of the following scenarios is/are ML-amenable?



When will my baby cry next?



Identify coins inserted into vending machine



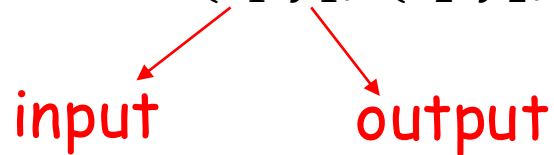
Convert handwriting to digital text

Three kinds of ML

1. Supervised learning

- Given (typically lots of) training data samples

$(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$


input output

- Goal:

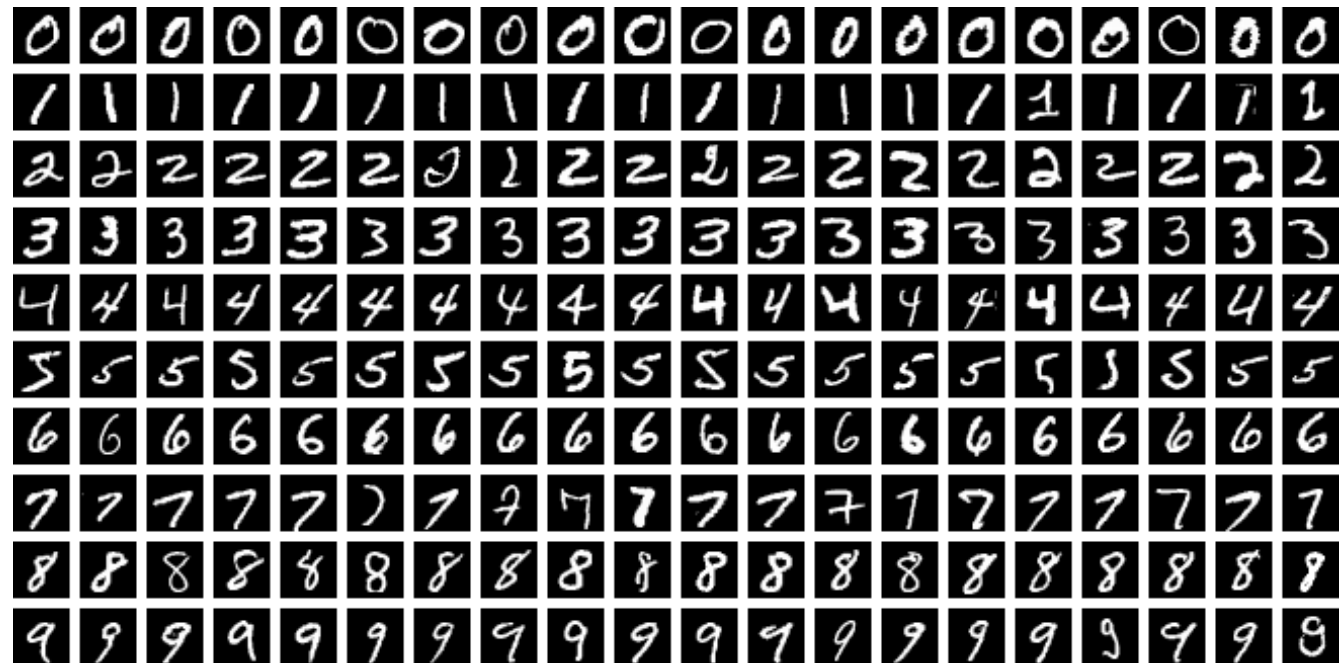
Predict output y corresponding to unseen input x (prediction)

Infer the nature of the relationship between input and output (inference)

most of this course

Example: handwritten digit recognition

x : an image of the form



(MNIST dataset)

y : number in image

Example: product marketing

x : (TV ad spend, radio ad spend, online ad spend)

y : sales

Example: loan processing

x : (age, salary, credit history)

Case 1: y : loan interest rate

Case 2: y : accept/reject loan application

An example we are all familiar with...

x : (incomplete) string of English text

y : *next* word in the string



Claude

Stanford
Alpaca



Gemini



deepseek



Grok

Three kinds of ML

2. Unsupervised learning

- Given (typically lots of) data samples of the form

$$x_1, x_2, \dots, x_n$$

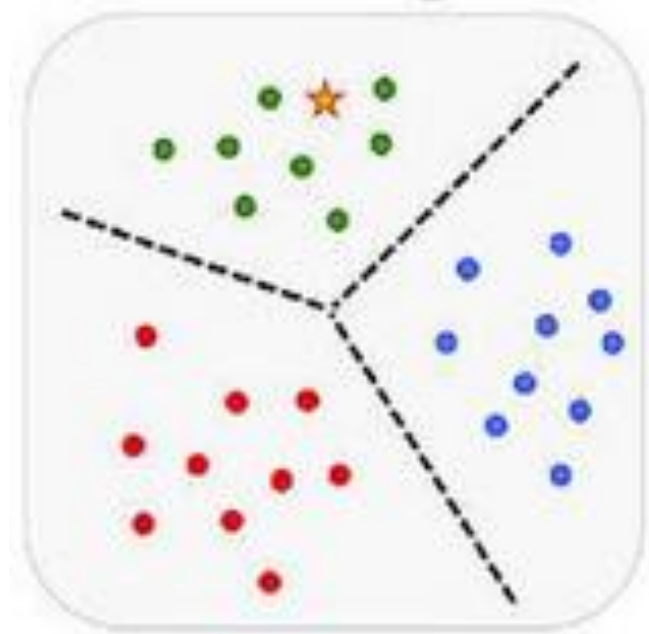
no "output"

- Goal:

Find patterns in the data

Eg: Clustering

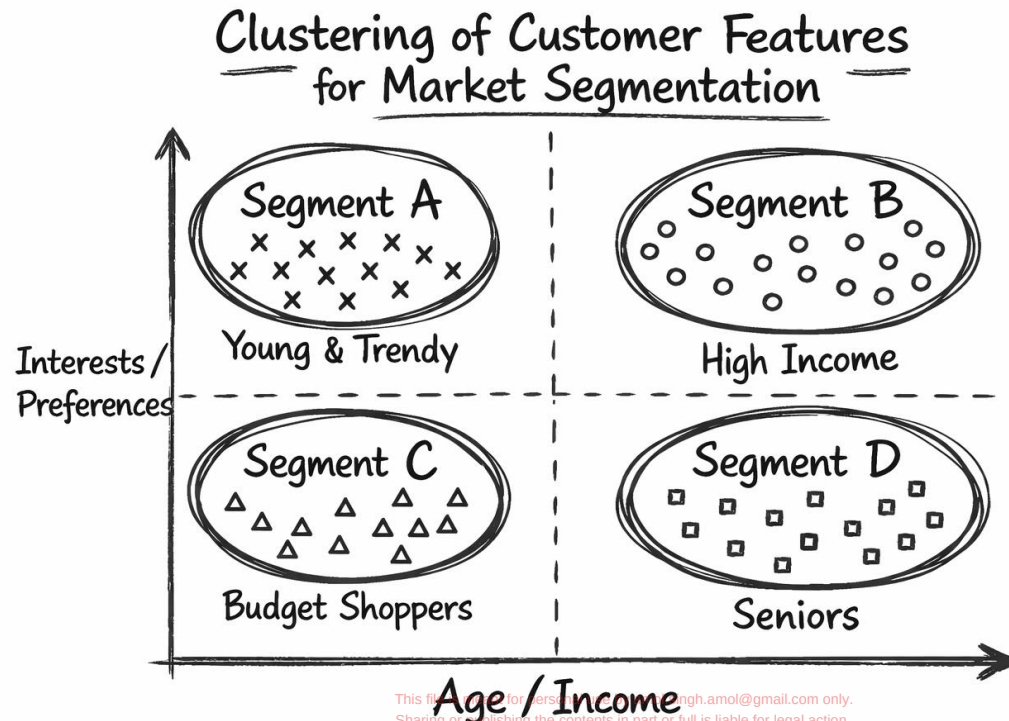
some coverage in this course



Example: market segmentation

$x = (\text{age, gender, pin code, spend history,....})$

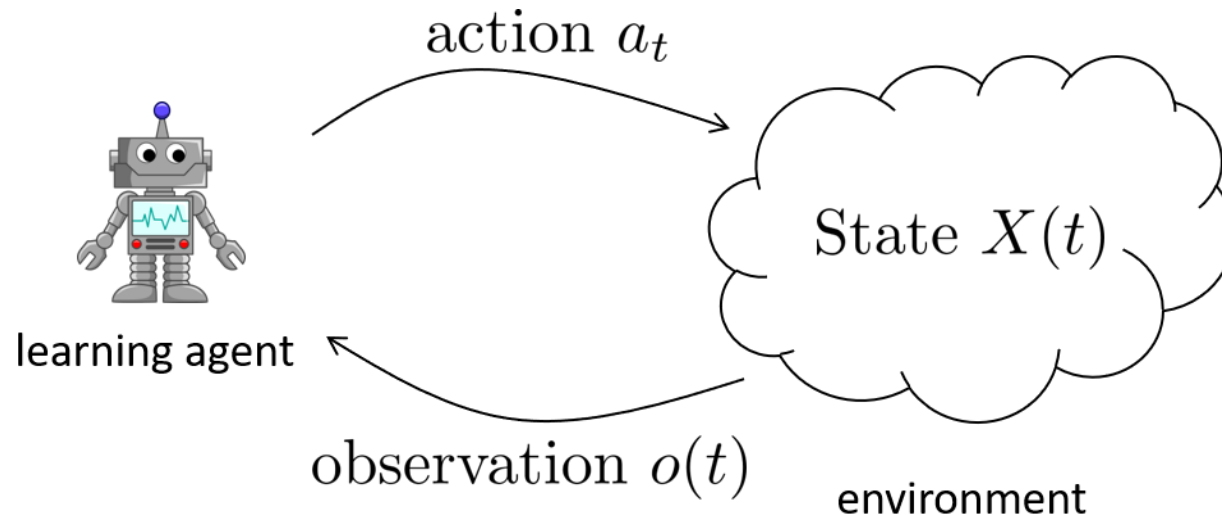
Goal: Segment customers into clusters (for marketing / product dev.)



Three kinds of ML

3. Reinforcement Learning

Learning in scenarios where a learning agent (a.k.a., algorithm) interacts with an (apriori) unknown environment sequentially



Env. state evolves based on current state and current action

Reward also depends on current state and action

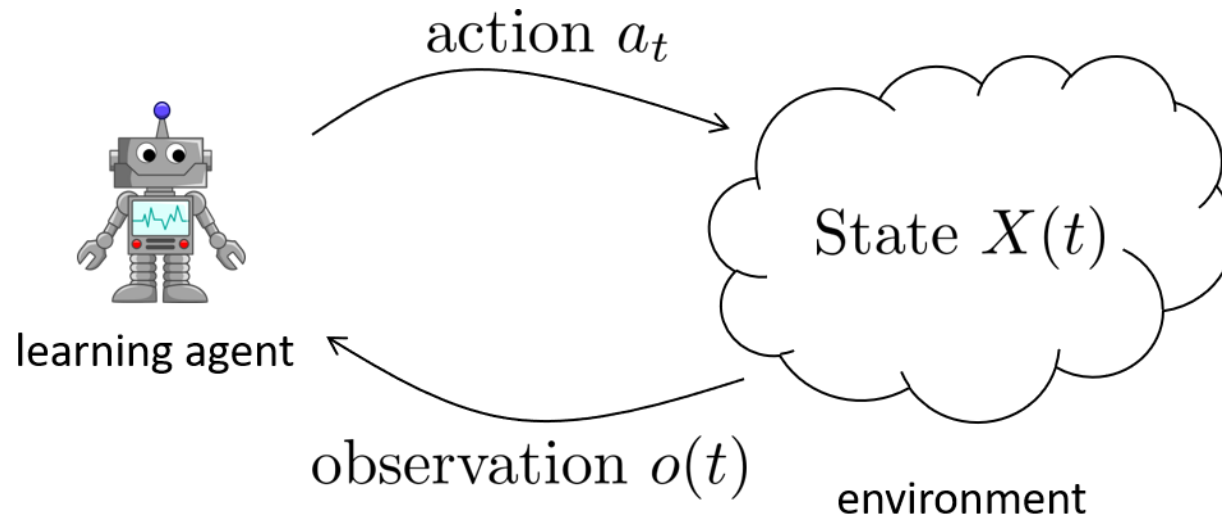
Present action influences future rewards

Q: How to optimally interact with the environment?

Three kinds of ML

3. Reinforcement Learning

Learning in scenarios where a learning agent (a.k.a., algorithm) interacts with an (a priori) unknown environment sequentially



Examples:

Autonomous driving
Robotic manipulation
Chess bots

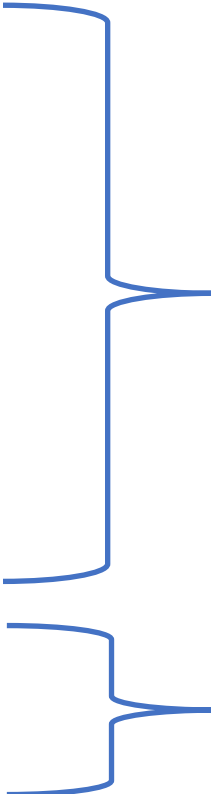
no coverage in this course

Three kinds of ML (summary)

1. Supervised Learning (most of this course)
2. Unsupervised learning (some of this course)
3. Reinforcement Learning (no coverage here – electives later?)

Course content

- Linear regression & Classification
- Model Selection & Regularization
- Cross-validation & Bootstrap
- Support Vector Machines
- Kernel methods
- Tree-based methods
- Deep neural networks
- Clustering
- Principal Components Analysis



supervised learning

The diagram uses blue curly braces to group the course topics. The first six topics (Linear regression & Classification, Model Selection & Regularization, Cross-validation & Bootstrap, Support Vector Machines, Kernel methods, and Tree-based methods) are grouped under the label 'supervised learning'. The last three topics (Deep neural networks, Clustering, and Principal Components Analysis) are grouped under the label 'unsupervised learning'.

unsupervised learning

Detailed course plan (first half)

Lecture #	Instructor	Material planned
1	JN	Introduction to AI & ML Supervised vs Unsupervised vs Reinforcement Learning
2	JN	Background in linear algebra
3	JN	Background in (unconstrained) optimization
4	JN	Linear regression -- ordinary least squares optimization, case studies
5	JN	Statistical & Geometric interpretation of least squares regression
6	JN	Model assessment & selection: subset selection, feature engineering
7	JN	Model assessment & selection: bootstrapping, in-sample methods, cross-validation
8		Introduction to constrained optimization: Primal, dual, KKT conditions
9	JN	Support vector machines: maximum margin classifier
10	JN	Support vector machines: soft margin classifier
11	JN	Kernel SVM
12	JN	Shrinkage methods for regression: Ridge regression & LASSO

Detailed course plan (second half)

Lecture #	Instructor	Material planned
13	SM	Probability refresher
14	SM	Gradient descent
15	SM	Classification: Logistic regression
16	SM	Classification: Gaussian discriminant analysis
17	SM	Classification: Naive Bayes
18	SM	Decision trees
19	SM	Decision trees (continued)
20	SM	Bagging, Random forest
21	SM	Boosting
22	SM	Unsupervised learning, K-means clustering
23	SM	Principal component analysis
24	SM	Principal component analysis (continued)

Textbook

Legally available for free download:

<https://www.statlearning.com>

Springer Texts in Statistics

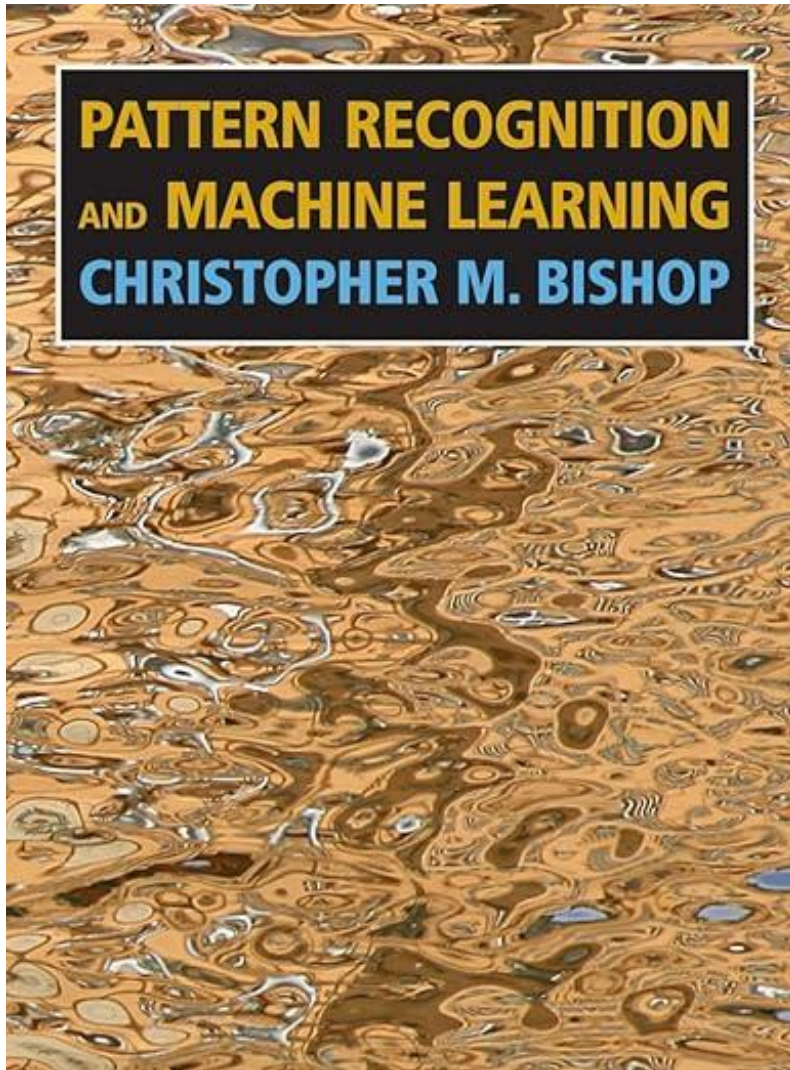
Gareth James · Daniela Witten · Trevor Hastie ·
Robert Tibshirani · Jonathan Taylor

An Introduction to Statistical Learning

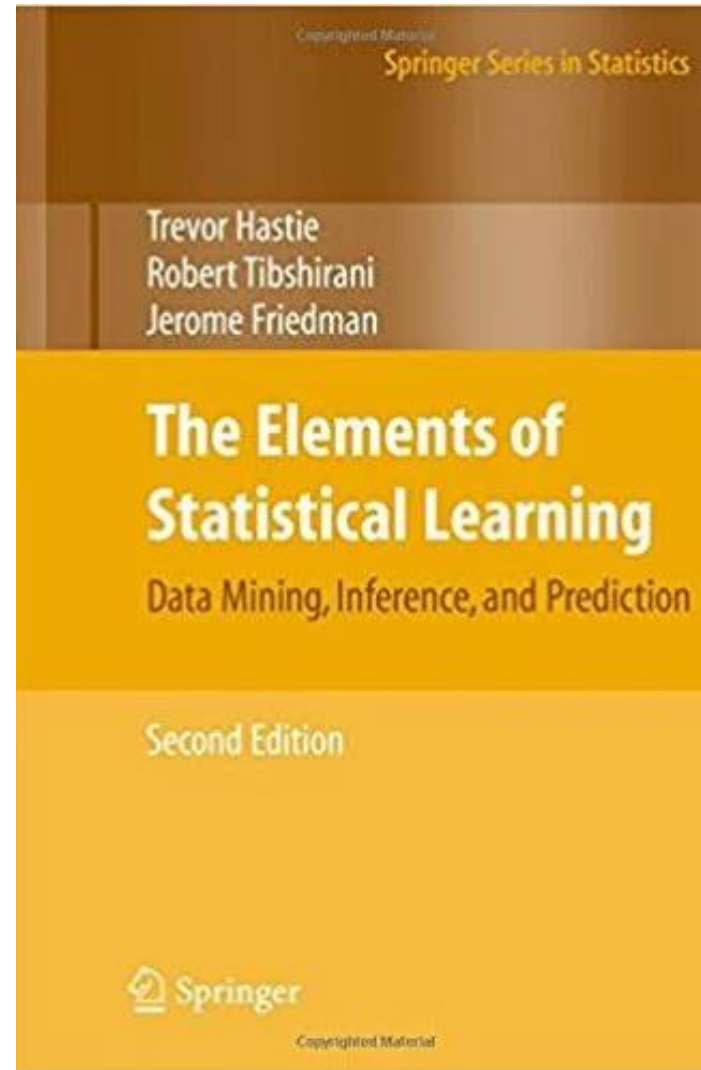
with Applications in Python

 Springer

Additional references (also legally downloadable for free)



[link](#)



[link](#)

This file is meant for personal use by amol.singh.amol@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Evaluation

- Projects (2) – 30% (one in each half of the semester)
- Mid-term exam – 20%
- End-term exam – 50%

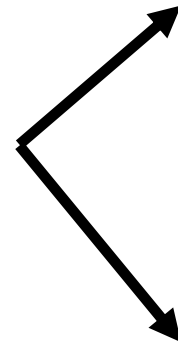
Course guidelines

- Attendance policy: 80% minimum attendance mandatory
- Attending lectures live is ***strongly encouraged***
- Expectation is that you spend 3 hours per week on this course (not counting live classes and tutorials)
- Passing the course requires scoring $> 40\%$ on the end-semester exam
- End-semester exam in person at IIT Bombay
- Gyan: Limit use of Gen-AI for projects, exam prep as well as coding exercises

Back to supervised learning

➤ Given: **training data**

$(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$



prediction

predict y corresponding to unseen x

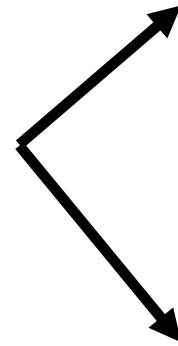
inference

infer nature of the input-output relationship

Back to supervised learning

➤ Given: **training data**

$(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$



prediction

predict y corresponding to unseen x

inference

infer nature of the input-output relationship

Ex: Does a certain input feature influence output?

Does gender influence loan default probability?

Back to supervised learning

➤ Given: **training data**

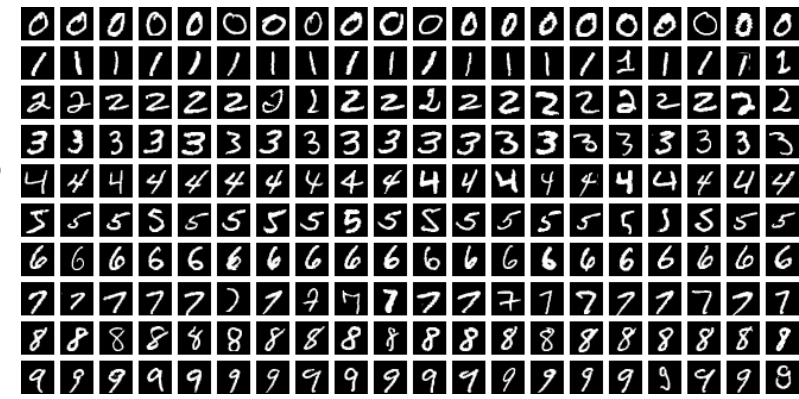
$(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$

lie in $\mathbb{R}^d$

Example:

components called inputs/predictors/features/independent variables

Q: What is the input in the MNIST digit recognition task?



Back to supervised learning

➤ Given: **training data**

$(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$



one-dimensional for now
called output/label/dependent variable

Back to supervised learning

➤ Given: **training data**

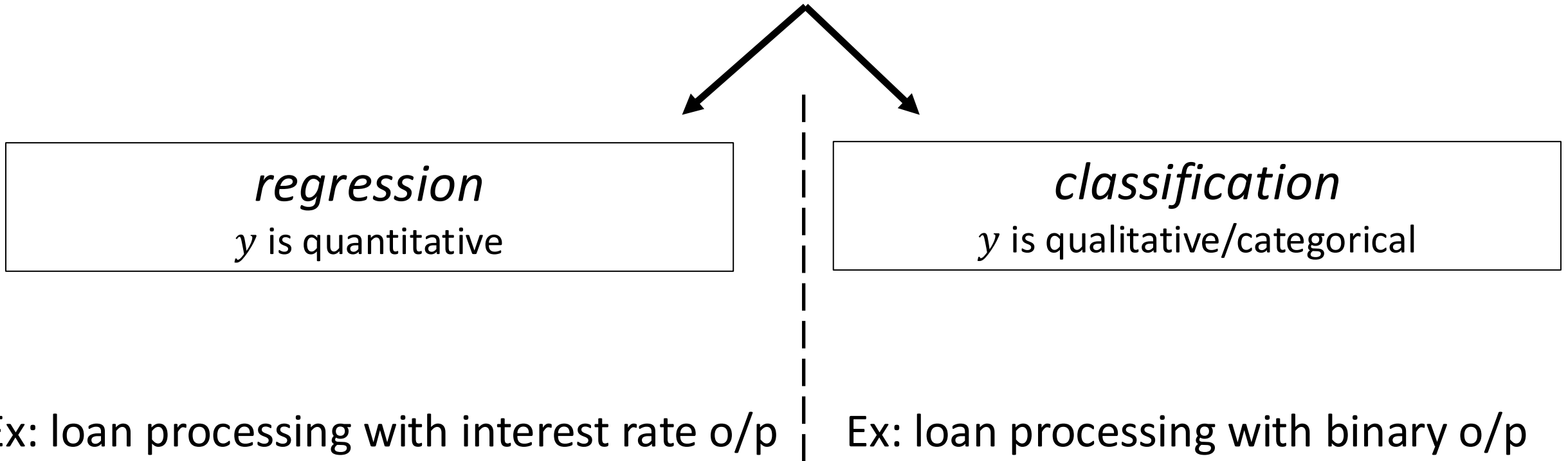
$$(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$$

Variables (input and output) can be of different *types*:

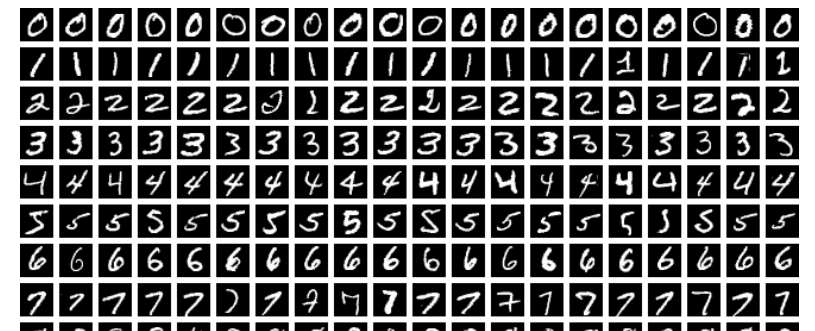
- Quantitative (real valued)
- Qualitative/categorical/discrete

(e.g.: 0/1, Cat/Dog/Monkey)

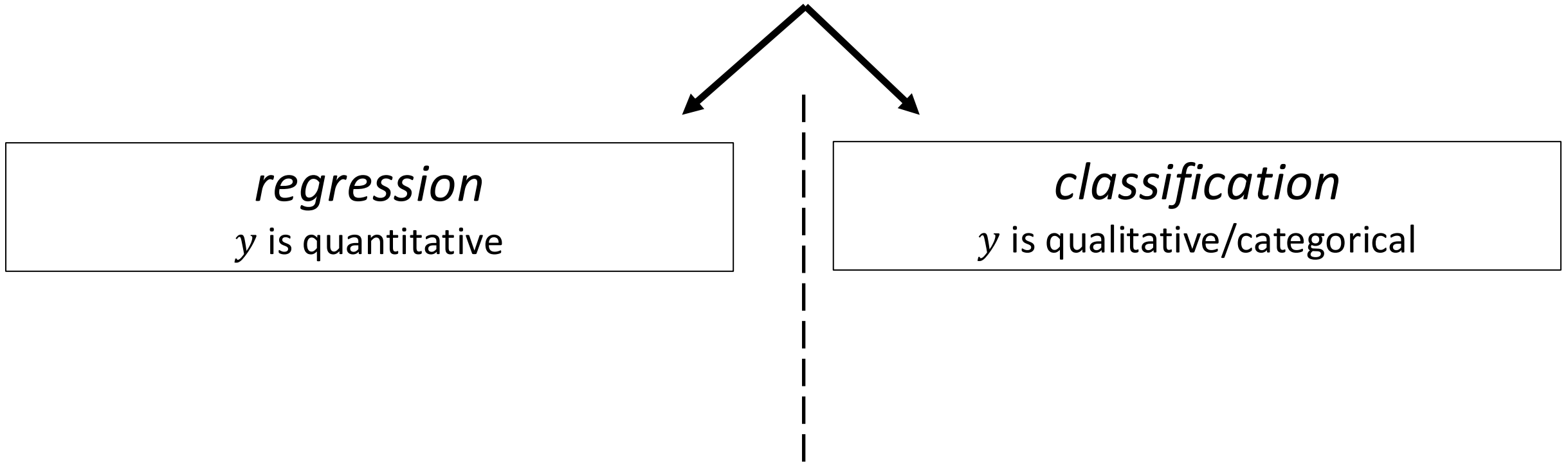
Prediction in supervised ML



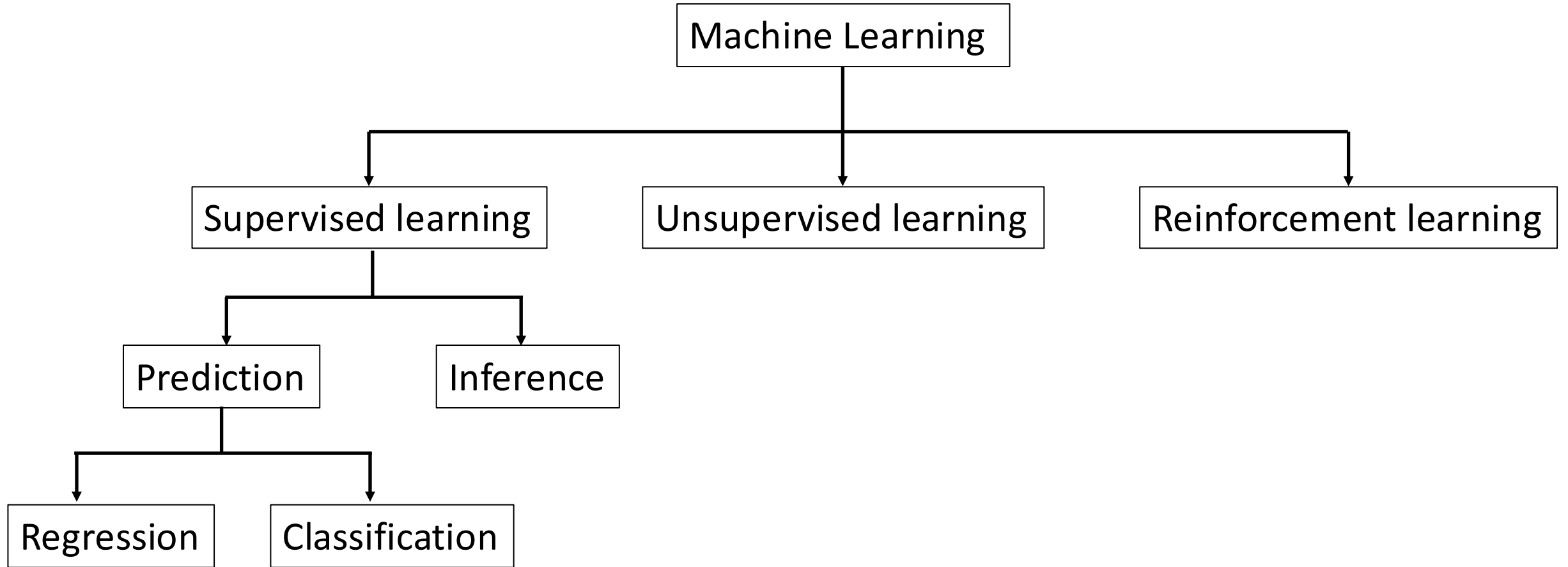
Q: Category of the MNIST digit recognition task?

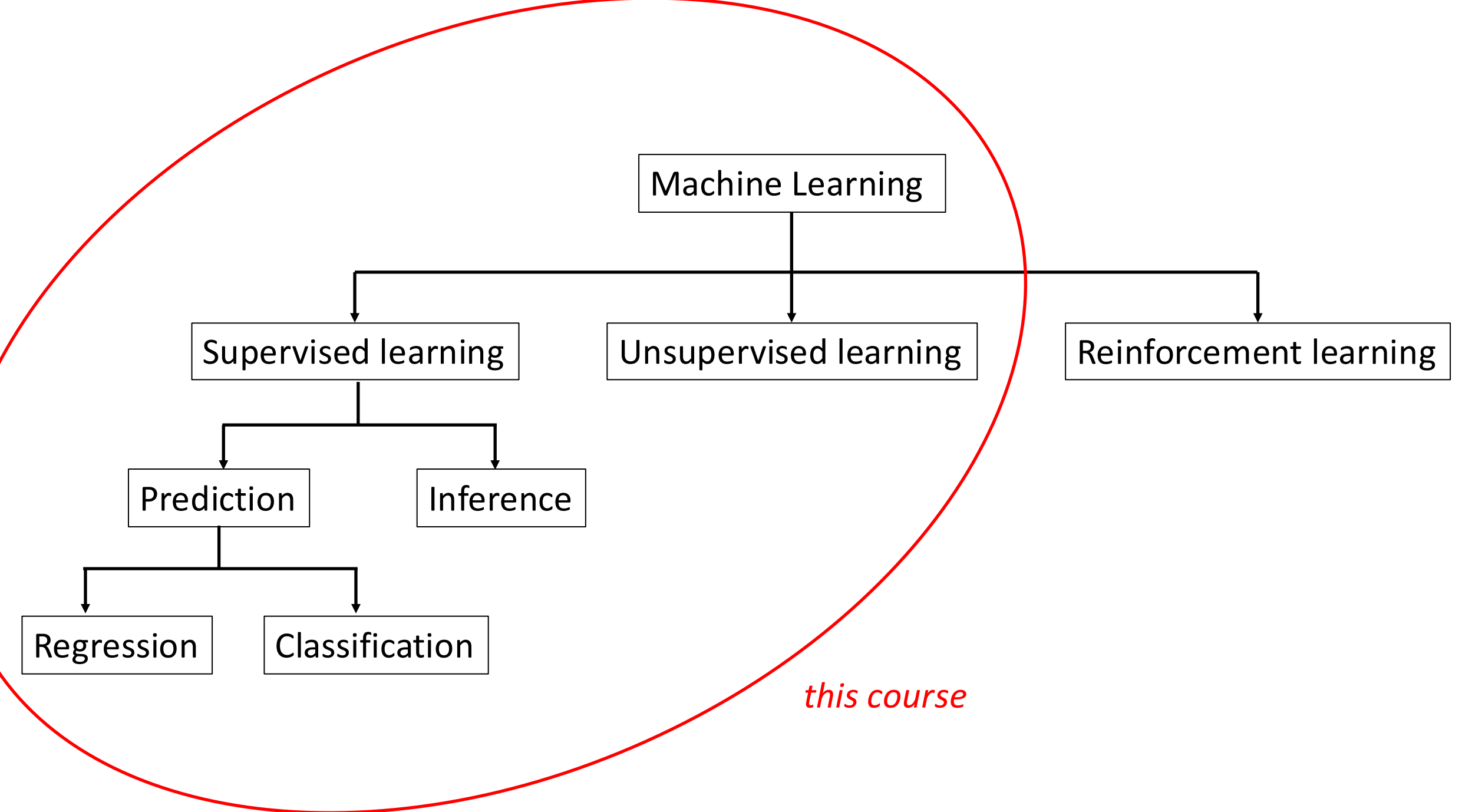


Prediction in supervised ML



- Different techniques for regression and classification tasks
- Will study both as part of this course





Relation of ML to other fields



** courtesy: Prof. Amit Sethi*

Some interesting videos



The Epistemology of Deep Learning - Yann LeCun